

Vikash Kumar

Machine Learning & AI Research Engineer

📍 East Delhi, India

☎ +91 92050 89152 | ✉ vktofly@gmail.com

🌐 linkedin.com/in/vktofly | 🐙 github.com/vktofly | 🌐 vktofly.github.io

SUMMARY

Machine Learning & AI Research Engineer with a deep mathematical foundation in multivariate calculus, numerical linear algebra, probability theory, and convex optimization. Experienced in developing custom machine learning architectures, loss function formulations, and autonomous multi-agent reasoning systems. Passionate about foundational model mechanics, algorithmic efficiency, and building robust, mathematically grounded AI solutions from first principles.

MATHEMATICAL & TECHNICAL FOUNDATIONS

Mathematical Foundations: Numerical Linear Algebra, Multivariate Calculus, Probability Theory & Stochastic Processes, Convex & Non-Convex Optimization, Statistical Inference

Core AI & Deep Learning: Transformer Architectures, LLM Alignment (RLHF, DPO, LoRA), Multi-Agent Systems (LangGraph, CrewAI), RAG, Attention Mechanisms, Custom Loss Functions

Infrastructure & Frameworks: Python (Async, PyTorch, TensorFlow, NumPy, SciPy), C++, CUDA, GCP (Google Cloud), Docker, Kubernetes, SQL, Git

Applied Engineering: Algorithmic Complexity, Cost & Compute Optimization, Model Evaluation (LangSmith, DeepEval), Distributed Data Processing

EXPERIENCE

Mar 2025 – Present

Remote

Founder & Principal Engineer | *Mono Pvt. Ltd.*

- Architected self-hosted agentic workflows (LangChain, n8n, Python, Supabase) and managed client projects via Kanban, ensuring on-time delivery of custom LLM solutions, reducing manual processing time by 34%.

Mar 2024 – Jan 2025

Noida, India

AI Software Engineer | *Techype Pvt. Ltd.*

- Built a talent acquisition platform via AI-assisted development (Cursor, Claude), streamlining candidate sourcing to reduce time-to-hire by 40%.
- Engineered full-stack features from ideation to deployment using Agile methodologies, collaborating cross-functionally to align technical deliverables with business goals.

Jul 2024 – Jan 2025

New Delhi, India

IT Manager & Operations | *Aparc Healthcare*

- Directed IT infrastructure across a multi-department facility, ensuring 99%+ uptime while providing technical leadership and bridging technology gaps.
- Automated and optimized patient data management and scheduling workflows using internal tools, improving daily operational efficiency by [X]%.

RESEARCH & PROJECTS

KV-Cache Compression & Optimization Prototype [GitHub]

- Engineered a mathematical simulation in Python demonstrating token-eviction strategies and PagedAttention logic to reduce memory bottlenecks during long-context LLM generation.
- Solved significant LLM inference constraints by modeling computational complexity, reducing simulated VRAM usage by [X]%.

Direct Preference Optimization (DPO) from Scratch [GitHub]

- Implemented the mathematical DPO loss function in PyTorch, fine-tuning a local Llama-3-8B model to align outputs without relying on traditional RLHF frameworks.
- Orchestrated advanced fine-tuning architectures to demonstrate mastery of modern model alignment mathematics, improving response coherence by [X]%.

Custom GPU Kernel via OpenAI Triton (Attention) [\[GitHub\]](#)

- Authored a custom Sliding Window Attention mechanism using Triton, benchmarking its memory footprint and throughput against standard PyTorch implementations.
- Optimized VRAM usage and memory bandwidth, increasing inference throughput by [X]% and proving deep understanding of hardware constraints.

EDUCATION

2020 – 2023	●	M.Tech in AI/ML (online, self paced) <i>IIT & MIT OpenCourseWare</i>
2016 – 2020	●	B.Tech in Computer Science and Engineering <i>GB Pant Govt. Engineering College (GGSIPU) – 74%</i>
2012 – 2015	●	12th & 10th (74%) <i>CBSE Board – Advanced Mathematics and Sciences (PCM)</i>

CERTIFICATIONS & HONORS

Certifications: Master of Artificial Intelligence (UPenn), Mathematics for Machine Learning (Coursera), Transformer Models & BERT (Google Cloud), SARS-CoV-2 Protein Modeling & Drug Docking (Coursera)

Languages: English (Full Professional), Hindi (Native), German (Elementary)